Formal Guarantees of Error-bounded Lossy Compression

Tripti Agarwal, Ganesh Gopalakrishnan

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Large Scientific Data

Today's scientific research relies on **simulations and instruments** that **generate extremely large datasets** for processing and analysis.

Image source: https://szcompressor.org/tabs/application/#weather

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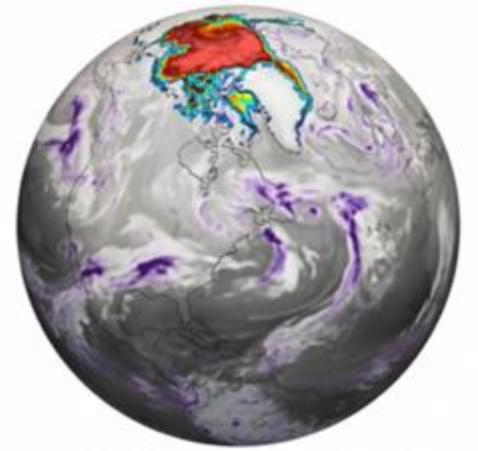

- **Climate Simulation**
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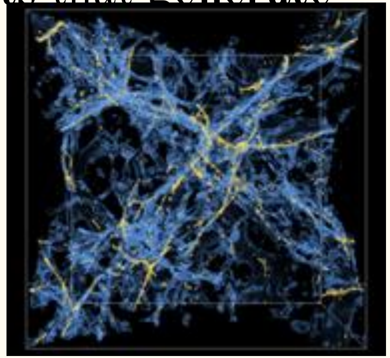

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How to reduce the data without losing important information?

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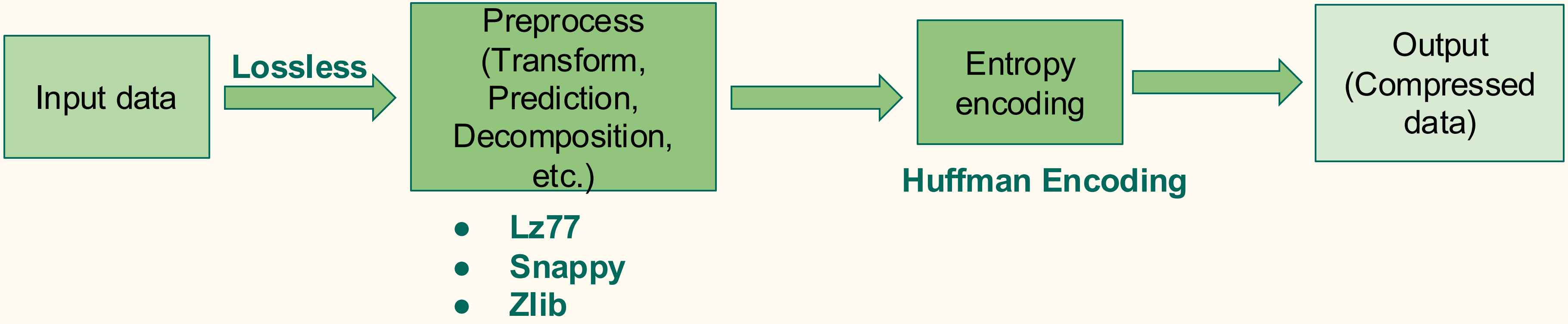

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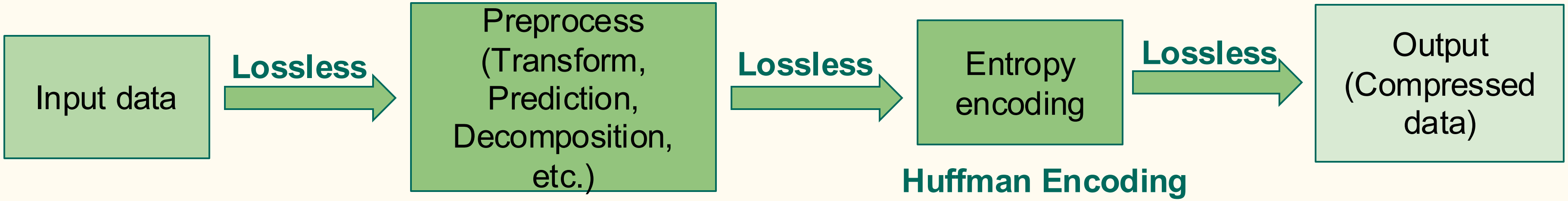
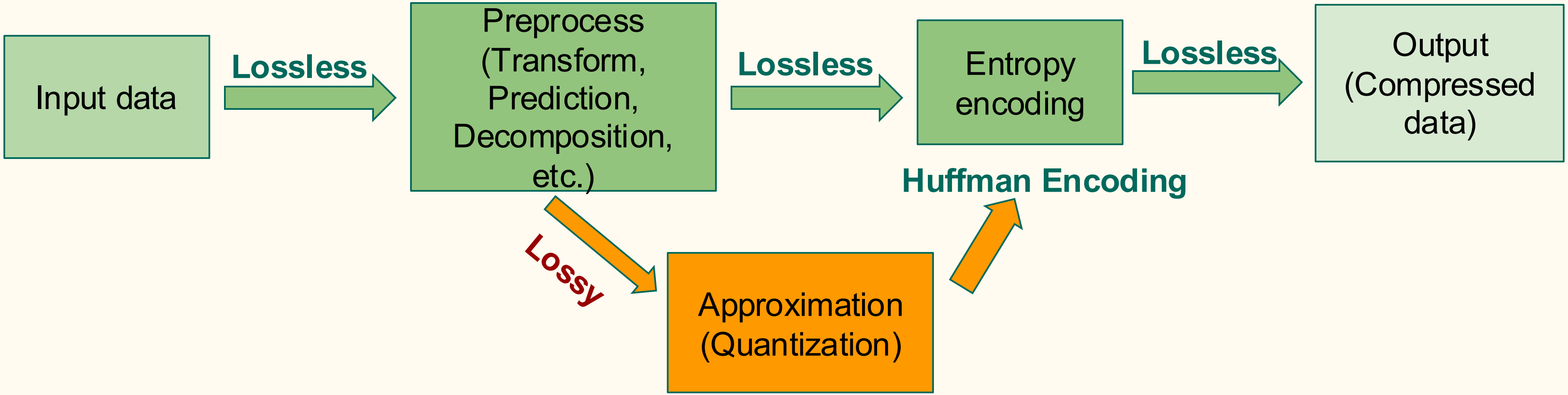


- **Lz77**
- **Snappy**
- **Zlib**

Lossless compression is limited to a ratio of 2 for scientific data

Lossy compression

Reducing data by more than ratio of 2



Compression error is introduced here

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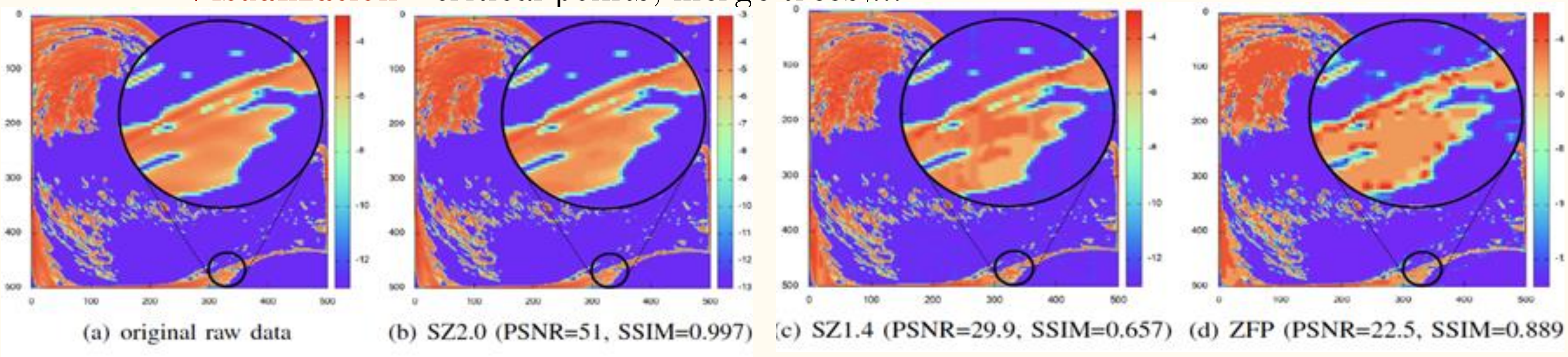

Hurricane CLOUDf: slice 50, CR=66.1

Image source: https://szcompressor.org/tabs/performance/

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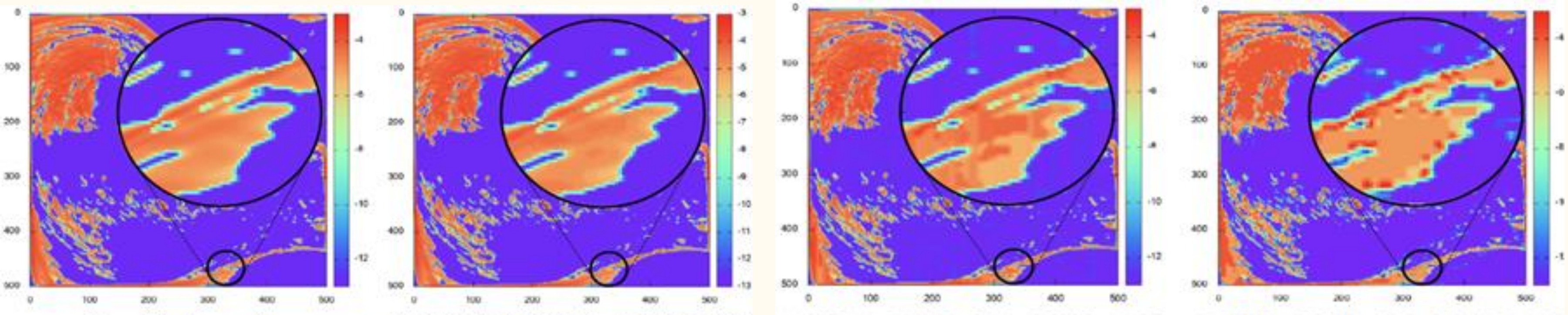
Quantitative assessment of lossy compressor errors

- Error metrics (amount of error)
  - **Error bounds - point wise metrics**
  - RMSE
  - PSNR
  - Rate distortion

$\left.\begin{matrix} \\ \\ \\ \end{matrix}\right\}$ Statistical metrics



(a) original raw data (b) SZ2.0 (PSNR=51, SSIM=0.997) (c) SZ1.4 (PSNR=29.9, SSIM=0.657) (d) ZFP (PSNR=22.5, SSIM=0.889

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Error bounds calculation

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  - **Absolute error bound**
    - $E_a = | V - V' |$
  - **Relative error bound**
    - $E_r = \frac{| V - V' |}{V}$

where, E: error bound, V: initial dataset, V': decompressed dataset

[1] https://szcompressor.org/
[2] https://computing.llnl.gov/projects/zfp

Lossy compression error - effect on data

- **Ideally**, we can **formalize the compression error** and propagate them in the simulation equations
- **In reality**, it is **very difficult**
- **Lack** a profound **understanding of lossy compression errors** effects

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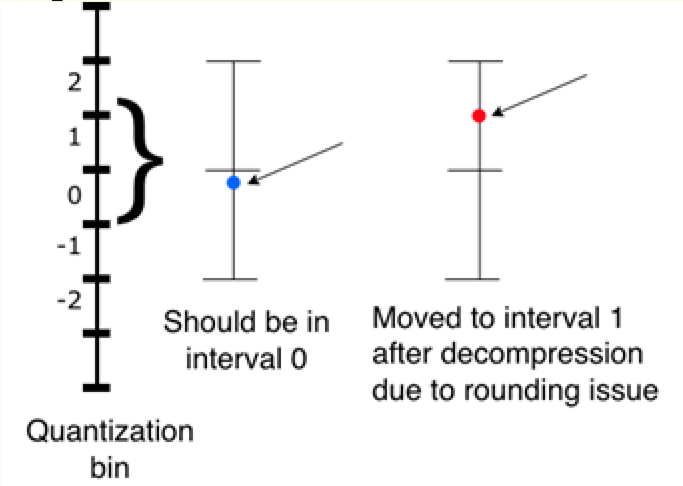

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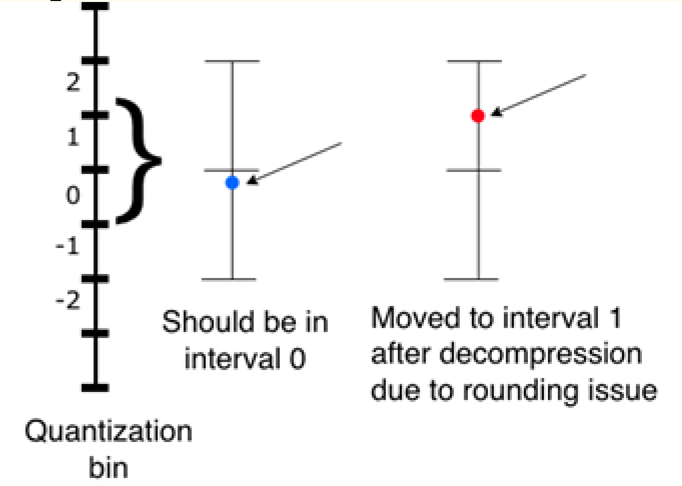

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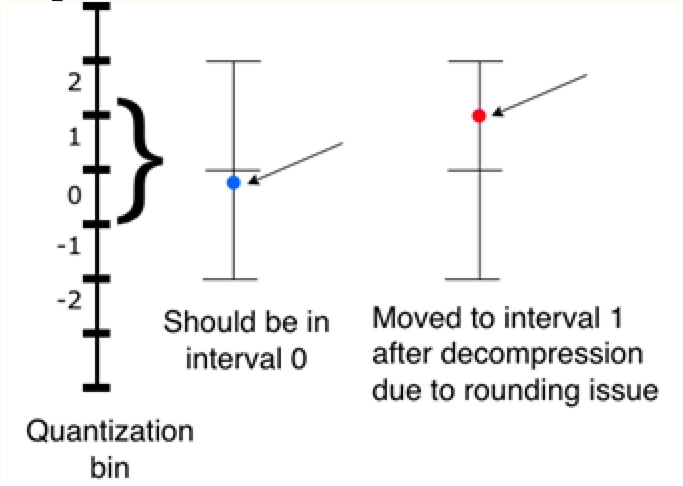
- **INF/ NaN** : propagate during computation
- **Denormals**: lower precision, highly sensitive to rounding



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- Specification-level guarantees:
  - Define contracts:
    - $| V - V' | \leq E$ for all finite V
    - Quantization must be monotonic, bounded and terminating
- Use **Verus**[1] and **express pre/postconditions and invariants for quantization kernels**
- Verus is a Rust based verification language

[1] https://github.com/verus-lang/verus

Race condition elimination

- Why race condition matters?
  - Parallel compressors (OpenMP/CUDA) are fast but risk data races in shared states
  - **Race conditions → Non-deterministic behavior → break correctness guarantees**

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  - **Race conditions → Non-deterministic behavior → break correctness guarantees**
- How using Verus is helpful
  - **Ownership and borrowing models**
    - Each thread operates on disjoint memory segments
  - **Permission based reasoning**
    - No two threads write in the same memory cell simultaneously

Visualization - Topology based quality metrics

- Topological descriptors

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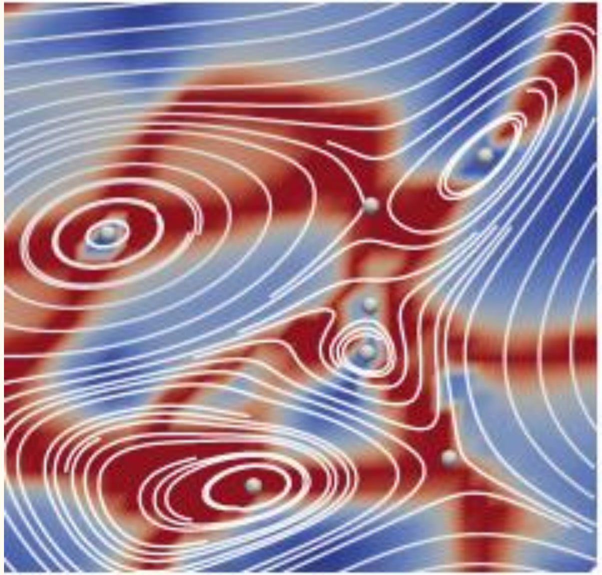
  - **Critical points in vector or scalar fields**



Streamlines and critical points

Image source: X. Liang, S. Di, F. Cappello, M. Raj, C. Liu, K. Ono, Z. Chen, T. Peterka,and H. Guo, “Toward featurepreserving vector field compression,” IEEE Transactions on Visualization and Computer Graphics, 2022.

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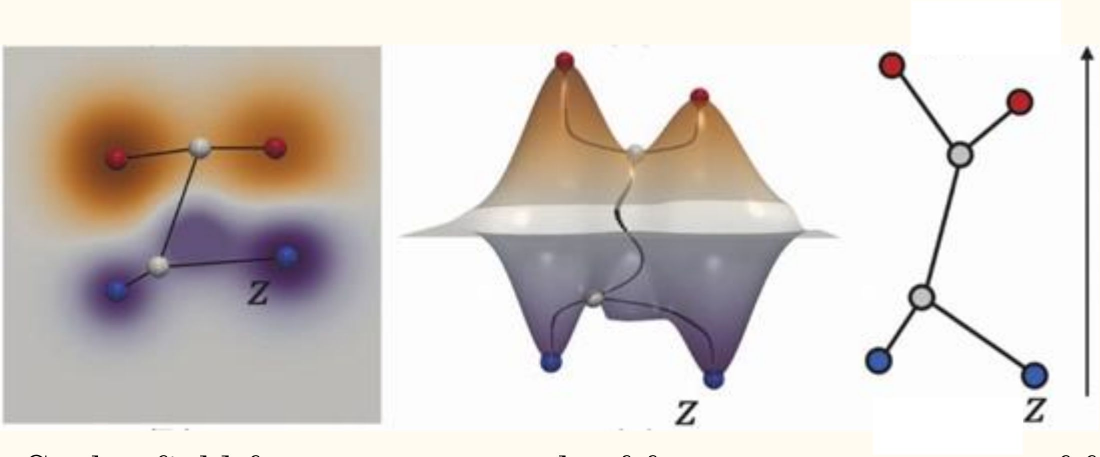
  - **Contour trees**



Scalar field f   graph of f   contour tree of f

Image source: Yan et al. “TopoSZ: Preserving Topology in Error-Bounded Lossy Compression, 2023”

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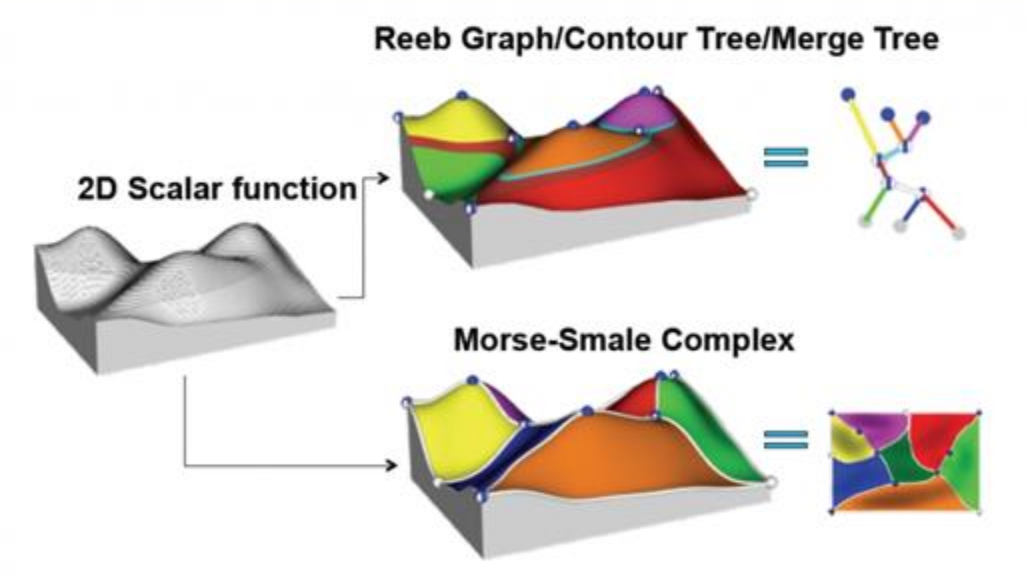
  - **Merge trees and Morse-smale**



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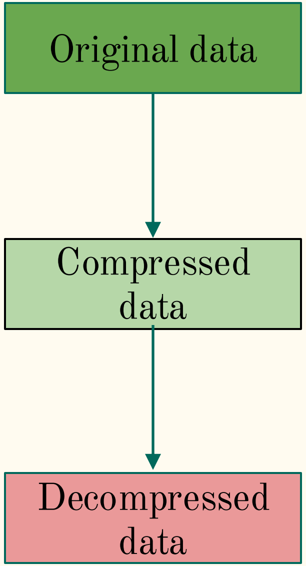
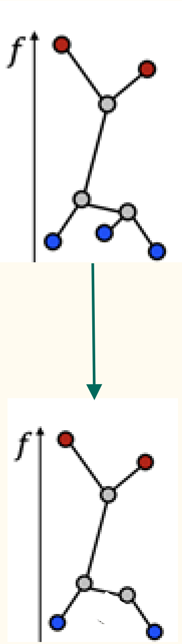
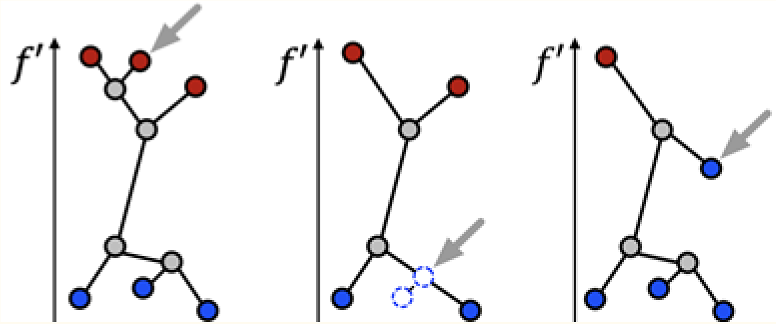
False cases





original contour tree

contour tree constructed by decompressed data



False Positive (FP) ✖

False Negative (FN) ✖

False Type (FT) ✖

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Solution

- Topology community is well aware of the problem
- Different solutions exists:
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- **But all these solutions have multiple iterations or are bottlenecked by the topological descriptor construction**

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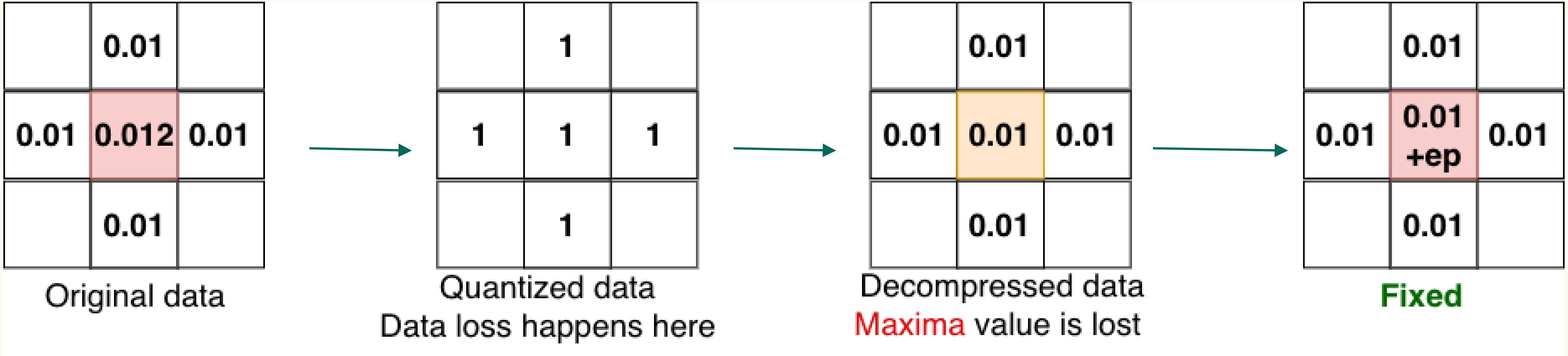

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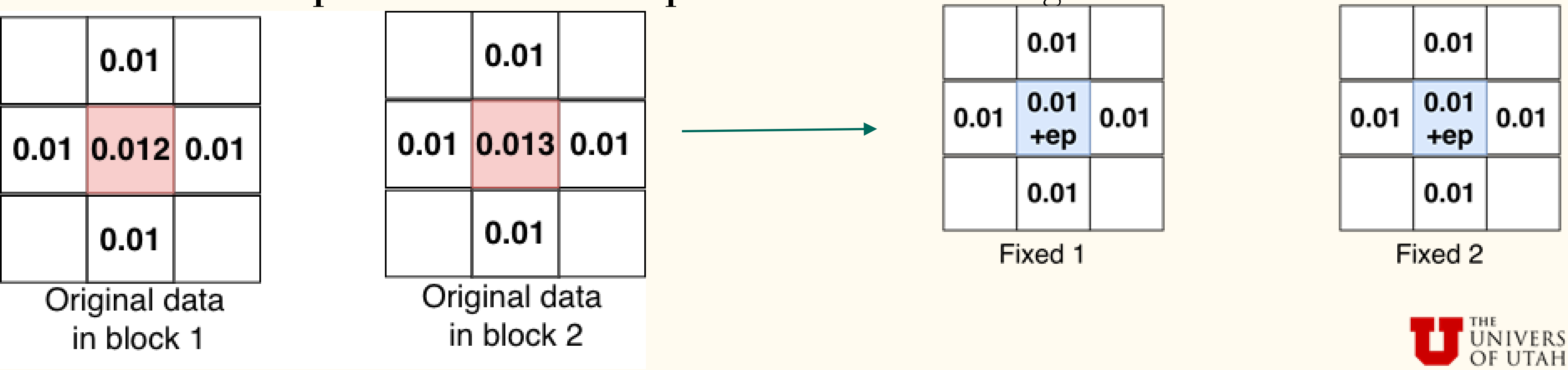

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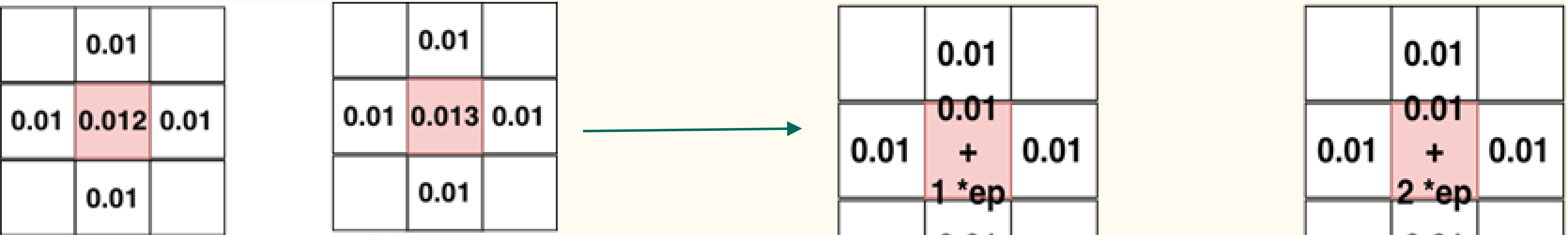

Original data in block 1

Original data in block 2

Fixed 1 with relative position

Fixed 2 with relative position

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  - **Single-pass topology preservation** is fast and reliable
- Together they enable **formally correct, deterministic and efficient lossy compression** for scientific workflows

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Thank you!

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